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How Can Genomics Help or Hinder Wildlife Conservation?

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Keywords

genetic structure, heterozygosity, kinship, genetic load, adaptation, management

Abstract

Genomic data are becoming increasingly affordable and easy to collect, and new tools for their analysis are appearing rapidly. Conservation biologists are interested in using this information to assist in management and planning but are typically limited financially and by the lack of genomic resources available for non-model taxa. It is therefore important to be aware of the pitfalls as well as the benefits of applying genomic approaches. Here, we highlight recent methods aimed at standardizing population assessments of genetic variation, inbreeding, and forms of genetic load and methods that help identify past and ongoing patterns of genetic interchange between populations, including those subjected to recent disturbance. We emphasize challenges in applying some of these methods and the need for adequate bioinformatic support. We also consider the promises and challenges of applying genomic approaches to understand adaptive changes in natural populations to predict their future adaptive capacity.

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1. INTRODUCTION

Genetic load: loss of fitness, relative to an optimal genotype, in a population or individual due to deleterious mutations

Conservation genetics (CG) is now an established area of research focused on providing conservation biologists with information on genetic variation, gene flow, population structure (contemporary and historic), viable population size, inbreeding, and pedigree analysis (summarized in 1, 2). Many recent reviews in this area have covered a range of topics, including the development and use of reference genomes (3, 4) and how genomics can be used within the context of climate change adaptation (5–7), decisions around species translocations (8), and applications to management (9). These highlight the excitement around using genomics in different contexts, but practical applications with documented benefits remain limited.

Although genomic approaches are powerful, their application has important pitfalls. Unlike traditional genetic markers, which are relatively tractable to manual quality control, genomic data (see the sidebar titled What Are Genomic Data?) are vast, and quality control is challenging. Many sophisticated analytical methods have been developed with specific assumptions and idealized data and are often tested on well-studied organisms with excellent genomic resources (including humans). Extension of these analytical methods to non-model species may not be clear, or there may be misconceptions around their appropriate application. Many characteristics of species of conservation concern impose limits on the types of analyses that can be performed.

In this review, we highlight advances and challenges in some key applications of genomic data to conservation issues. We outline the novel insights that genomics could provide for understanding long-standing issues in CG, covering population structure as well as gene flow, patterns of genetic variation relevant to inbreeding, genetic drift, evolvability, and genetic load. Genomic approaches provide a much more dynamic picture of past and present evolutionary processes affecting population genetic structure and genetic variability—informing managers about conservation objectives without being prescriptive. Nevertheless, important challenges remain around suitable data sources and analysis packages in CG. Genomic approaches to study past and future adaptive processes in threatened species are particularly fraught with problems and require caution.

We briefly consider some applications of genomic approaches to conservation management. The success of these applications depends on how a species' biology is factored into the study goals, design, and budget. Careful decision making will ensure that money spent producing and analyzing genomic data leads to meaningful insights that are relatively unbiased by bioinformatic handling, analytical artefacts, or other factors.

WHAT ARE GENOMIC DATA?

Throughout this review we refer to genomic data. This term covers any type of empirical DNA sequence data that encompasses a large and representative subset of regions across the genome. Such data can be generated by RADseq (restriction-associated digestion sequencing), pool-seq (sequencing of pooled individuals), whole-genome resequencing (high or low coverage), exome sequencing, SNP (single-nucleotide polymorphism) chip, or sundry other methods. Although there are important differences between these data types, we refer to them collectively as genomic data, which share fundamental characteristics that set them apart from traditional genetic markers. The latter include COI (cytochrome oxidase 1) mitochondrial DNA loci and microsatellites, which typically have lower resolution for identifying the effects of selection and establishing population processes. Genomic data involve large data sets, which provide much greater analytical power and are essential for a wide range of specific analyses, but also require specific skill sets for filtering and analysis.

2. TO ASSEMBLE, OR NOT ASSEMBLE, A REFERENCE GENOME?

A reference genome describes the organization, structure, and gene content of a species' genome, helping place genetic sequences in their broader genomic context. The capacity to generate reference genomes has been enhanced recently by long-read sequencing technologies, powerful scaffolding approaches, sophisticated genome assembly programs, and community-created how-to guides. Previously, when a focal species lacked a reference genome, a genome of a closely related species has been used. However, increasing divergence between species increases the rate of miscalled genotypes and unmapped reads, which may bias estimates of genetic diversity and confound inferences of demographic and adaptive processes (e.g., 10–13). Some excellent reviews on the applications of reference genomes in conservation already exist (e.g., 3, 14, 15). We also note that increasing capacity for genome assembly is leading to an era of pangenomes. Although useful for studying the structural variants involved in adaptation (e.g., 16–18), pangenome-level studies are within the scope of only the most well-funded conservation projects in the immediate future, and particularly for species with large genomes.

Genome-wide association (GWA): identification of genetic markers that are statistically associated with phenotypes for a focal trait

Genotype–environment association (GEA): identification of genetic markers that are statistically associated with environmental gradients

High-quality reference genomes that are contiguous (not fragmented), accurate (not misassembled), and complete (contain all genes) are most valuable for characterizing the genetic architecture of key traits and adaptation in genome-wide associations (GWAs) or genotype–environment associations (GEAs) (Section 7) (see 19, 20). For other population genetic analyses, a draft reference genome or no genome may be sufficient (**Supplemental Table 1**). Until very recently, it was necessary to combine an Illumina whole-genome short-read library with at least one long-read sequencing technology when assembling a reference genome. However, the recent advent of the PacBio Revio system could revolutionize genome assembly by providing high output (up to 90 Gb per SMRT cell) of very high-quality long reads (99.95% accurate, Q33) (21). This capacity of the PacBio Revio system may negate the need to combine short- and long-read technologies. This method may be less cost-effective for organisms with small genomes (<1 Gb) relative to combining Oxford Nanopore Technology with Illumina (**Supplemental Figure 1**). However, for genomes approaching 9 Gb, it may even be possible to obtain a usable reference for ~USD10,000 (**Supplemental Figure 1**), but this will depend on other properties of the genome.

Supplemental Material >

Sequencing costs continue to fall, but the funds required to assemble a reference genome may still be limiting in many practical situations. In addition to sequencing costs, there also remains the nontrivial need to engage a proficient bioinformatician to conduct the genome assembly. There are different avenues to gauge feasibility and reduce costs. A pilot whole-genome Illumina data set can be useful for estimating genome size, heterozygosity, and repeat content, for example, by using GenomeScope (22), before further funds are invested in sequencing. Heterozygosity and repeat content themselves can also downwardly bias genome size estimates (23), so if these parameters are large, required sequencing investment may be much higher than that based purely on the genome size alone. Costs can be further reduced by performing restriction-associated digestion sequencing (RADseq), sequencing pooled individuals (pool-seq; 24, 25), or using low-coverage whole-genome sequencing (26, 27). The limitations of these cost-saving strategies must be factored into a study's objectives.

3. THE POPULATION AS A HYPOTHESIS: NEW GENOMIC METHODS FOR ASSESSING STRUCTURE

Identifying groups of individuals with distinct evolutionary histories is a key aim of CG. These groupings are given various names, including populations, demes, or ESUs (evolutionarily significant units) and MUs (management units) (28). Here, we use the term population to describe (*a*) an evolutionarily distinct group, the biological reality, and (*b*) an aggregation of individuals to

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be filtered and analyzed together, a hypothesis of what constitutes a population. Analyses of genetic structure investigate patterns of genetic differentiation between hypothesized populations. Differentiation can be summarized by *K*, which describes the number of discrete partitions in a data set, though methods need not estimate *K* directly and may instead produce hypothesis-free visualizations of individual variation (e.g., principal component analysis, PCA). Although individual-based genetic patterns are useful (Section 5), broad-scale, group-based statements about genetic structure remain critical for developing management programs and for delineating groups for further analysis. Genomic data provide power for delineating these groups. This section highlights new genomic methods that can improve population delineation, as well as studies testing the effects of different filtering and parameterization choices on downstream inferences.

***K*:** number of effective populations or unique ancestral lineages present in a genotype data set; there is no true *K*, only a *K* best supported by the data, and hence *K* is a hypothesis, not a known constant

Many tools currently used for assessing structure in conservation genomics were developed before genomic data became widespread, such as pairwise F_{ST}, STRUCTURE (29), PCA, and discriminant analysis of principal components (DAPC) (30). Although these methods remain useful, they do not consider the genomic position of variants and instead either ignore positional information (e.g., PCA) or treat it as a nuisance to be overcome through averaging over or omitting linked sites (e.g., in STRUCTURE). Various methods developed for genomic data can now explore structure in even higher resolution by directly incorporating positional information and the structure of phased haplotypes. For instance, RADtag microhaplotypes can provide detailed individual-based co-ancestry inferences via fineRADstructure (31), even if no reference genome is available. For whole-genome data, linkage-based analysis confers greater analytical power but requires a means of phasing variants; these include the foundational ChromoPainter/fineSTRUCTURE approach (32) to emerging methods employing variational autoencoders (33). Variational autoencoders can also be used free of linkage assumptions as an alternative to PCA or DAPC (34). These nonlinear methods can visualize both recently evolved structure and deeper differentiation patterns, offering a promising escape from the curse of dimensionality, whereby fine-scale and broad-scale structures are difficult to plot in 2D space. With careful implementation, variational autoencoders and their analogs may become an optimal choice for visualizing large data sets with complex genetic patterns.

Despite increasingly sophisticated methods, filtering genomic data remains a source of bias and confusion. One recurrent issue involves filtering rules of thumb that have been inherited from the pre-genomics age, such as filtering by minor allele frequency. It has become standard practice to filter for minor allele frequency $\geq 5\%$, but this disproportionately removes sites most informative of recent evolutionary history (35); a minor allele count ≥ 3 may generally be best for assessing structure. Similar issues have been identified with Hardy–Weinberg equilibrium filters (36). Applying the above filters too stringently can lead to erroneous conclusions of admixture, likely due to both the importance of rare and nonequilibrium alleles for assessing structure and the increased uncertainty of clustering with fewer data (35). Finally, when some individuals have missing genotypes, these are often accounted for by using the average genotype at a locus. If missing data are prevalent or unevenly distributed among populations, this can lead to erroneous conclusions of admixture (37). Imputation of missing data is one way to avoid these issues, and various imputation methods have been developed for reference-aligned data [e.g., Beagle (38)] or unaligned data [e.g., polyRAD (39)].

Following filtering, numerous downstream dangers await. Although defining a specific *K* population number may be the goal, there may be no obvious choice for *K*, particularly when patterns are shaped by continuous spatial structure (40, 41) or migration (42). *K* is sometimes chosen based on prior expectations from geography or phenotype, and these inferences may then lead to unsubstantiated statements about extant populations or the hypothetical ancestral populations that produced them (43). For methods using PC axes to model population structure, the number of PC

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axes chosen as predictors should be carefully considered (44). For example, although any number of PC axes can be used as predictors of among-population differences in a discriminant analysis, only the first $K - 1$ PC axes are biologically informative (45). Selecting too many additional PC axes artificially can therefore inflate perceived population structure in DAPC (44). Analyses can also be biased when unequal numbers of individuals from each true cluster are analyzed, leading to small clusters being erroneously combined into larger ones (46) or being projected farther in PC space (47). Structure is also expected to vary along the genome (48) and across space (40); ignoring continuous spatial structure may produce erroneous estimates of many parameters used in CG (49). Recent applications of the spatially explicit conStruct (40) identified different K optima compared to a nonspatial K-means clustering approach (50, 51).

Conservation geneticists must consider what questions are being addressed with genomic data and the logic leading from result to interpretation. For instance, upon identifying individuals that make up a highly differentiated cluster, how is this differentiation to be treated? It may be a product of isolation but could also result from drift operating on a small or peripheral population. Historical isolation provides a more compelling argument than recent drift for conservation (52), and demographic models and simulations can help evaluate support for different scenarios that may have produced contemporary differentiation patterns (53, 54). Isolation of individuals may go undetected if migration has become restricted only recently, such as for populations in recently fragmented landscapes where drift has had insufficient time to operate (50). This may lead to incorrect inferences that human intervention is not needed to maintain connections among habitat fragments. Migration estimates between two populations will also be confounded if other unsampled populations are sources of gene flow (55). Migration patterns may be assessed more effectively from the distribution of close kin (Section 5), particularly in recently changed environments.

Adaptive potential: capacity for populations to respond to selection and evolve to future environmental changes

Runs of homozygosity (ROH): sections of the genome where an individual has identical genotypes at each chromosome, i.e., where no sites are heterozygous; indicative of inbreeding and often quantified with respect to the frequency of differently sized homozygous genomic regions (e.g., number of ROH ≥1 Mb in the genome of an individual, or fraction of the genome that exhibits ROH ≥1 Mb in size)

4. HETEROZYGOSITY IN INDIVIDUALS AND POPULATIONS: THERE IS NO SUCH THING AS AN UNFILTERED SNP

Heterozygosity is a cornerstone of CG, as it points to the adaptive potential of populations while also indicating levels of past inbreeding (2, 56). Whereas structure describes genetic variation among populations, heterozygosity describes variation among chromosomes sampled within a population (expected heterozygosity) or within individuals (observed heterozygosity). Comparisons of individual whole genomes typically report observed heterozygosity for each individual separately (57, 58). Other data types, such as RADseq, also can be used to visualize patterns of individual-level heterozygosity across genomes (**Figure 1**). This approach can indicate genomic regions where individuals covary in heterozygosity, such as around the centromeres at the center of each autosome (**Figure 1*a***) and those where individuals share runs of homozygosity (ROH) (**Figure 1*b***). Autosome-wide variation can also be summarized for ease of comparison across groups (**Figure 1*c***). While the autosomal variation in **Figure 1*a*** benefits from an excellent reference genome, similar plots may be produced by concatenating fragmented genomes, though these will lack insight into specific regions.

Focusing heterozygosity at the individual level also appears to evade several recently raised filtering issues. These advantages come from processing each individual as a separate population of $n = 1$. Firstly, as missing genotypes have been linked to various estimation biases in heterozygosity (59, 60), setting $n = 1$ allows all missing sites to be filtered without omitting sites called confidently in other individuals. Provided the sequence data are spread broadly across the genome, there is no requirement that the exact same set of sites be used for heterozygosity comparisons, though comparisons across different data types should be aware of specific biases, such as from allelic dropout (61). Secondly, $n = 1$ ensures that tri-allelic and tetra-allelic sites are not filtered, as diploid individuals have at most two alleles at a site, and filtering these sites will downwardly bias estimates

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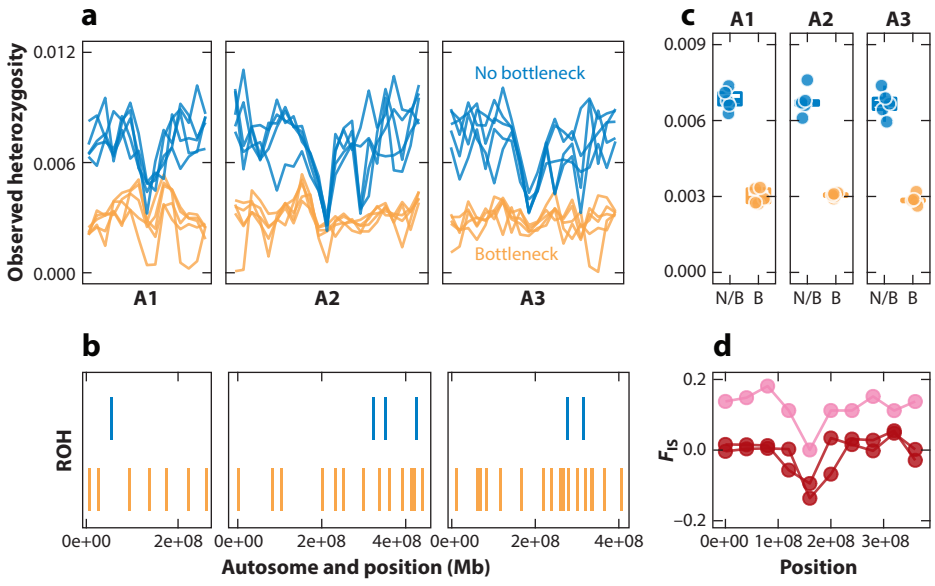

Figure 1

Visualizing autosomal heterozygosity. (*a–c*) Heterozygosity across three autosomes (A1, A2, and A3). Comparisons are of five individuals from a bottlenecked population and five from a non-bottlenecked population. (*a*) Heterozygosity for each individual, averaged across 20 Mbp. (*b*) ROH of length ≥1 Mbp found in ≥2 individuals in a population. (*c*) The data used in panel *a* as autosome-wide averages. (*d*) F_{IS} for two genetically similar populations ($n = 5$, *dark red*) and a sample taken across the two populations ($n = 5$, *rose*). All individuals were genotyped and filtered using a custom GATK/bcftools pipeline available at **https://github.com/t-ludovic-schmidt/autosomal_het**. Abbreviations: B, bottleneck; F_{IS}, inbreeding coefficient relative to population; N/B, no bottleneck; ROH, runs of homozygosity. Data: Author's own, double digest RADseq, *Aedes aegypti*.

(J. Sopniewski & R. Catullo, personal communication). It is important to clarify that processing each individual as a population of $n = 1$ takes place after individuals are genotyped. For genotyping, accuracy is greatest when all individuals are genotyped together, though combining strongly differentiated populations should be done cautiously (62). Workflows allowing for all individuals to be genotyped together, but filtered individually (e.g., by read depth, missing data), may be optimal for providing accurate heterozygosity estimates that are robust to cross-study comparisons (60).

Autosomal heterozygosity: genome-wide heterozygosity estimates derived from sites that have not been filtered by degree of polymorphism (i.e., derived from both monomorphic and polymorphic sites); sites on sex chromosomes typically should be omitted if positions are known

Notably, **Figure 1** plots data from all sites where the genotype has been confidently called. This includes monomorphic sites, which are often filtered during processing because they are uninformative for many analyses, but these sites are critical for assessing overall levels of variation. Metrics reporting variation at monomorphic sites and polymorphic sites (single-nucleotide polymorphisms, or SNPs) have been termed autosomal heterozygosity, as opposed to SNP heterozygosity, which considers only polymorphic sites and is biased upward as sample sizes decrease (60). With the right filtering, SNP heterozygosity can be used for comparisons across populations within a study, such as in assessing hybridization (63), but confers no advantage over autosomal heterozygosity for this. Autosomal heterozygosity is required for comparisons across studies and taxa. Because all SNPs are by definition filtered on the presence of polymorphism, there is no such thing as an unfiltered SNP. For this reason, a SNP data set cannot be used to assess total levels of genetic variation no matter how large the sample, as the number of invariant sites will be unknown.

Heterozygosity also varies across the genome, and these patterns can be shaped by neutral and adaptive processes. A key conservation application is to use these patterns to investigate

inbreeding, a catastrophic problem for species conservation. Here, we are interested in ROH (**Figure 1*b***), where the size and number of ROH are used to infer the age and severity of recent inbreeding events (54, 64). A small number of large ROH is typically a sign of recent inbreeding; eventually these recombine into many smaller ROH (65, 66). Distributions of ROH appear to be affected more by recent inbreeding than by recombination or selection (67). ROH are thus strongly linked to fitness, as shown by a study of Soay sheep (*Ovis aries*), in which a 10% increase in autosomal ROH corresponded to a 60% reduction in survival (68). Assessing ROH requires a reference genome with large scaffolds or contigs, but methods can incorporate recombination rate data to predict homozygosity across unsequenced regions (69).

SNP heterozygosity: genome-wide heterozygosity estimates derived from a set of sites known to be polymorphic (e.g., from a SNP data set)

Wahlund effect: observation of lower observed heterozygosity relative to expected heterozygosity as a result of population subdivision

Comparisons between observed (H_O) and expected (H_E) heterozygosity provide another means of investigating inbreeding, typically through the derived inbreeding parameter, F_{IS} (**Figure 1*d***). Investigating F_{IS} requires processing $n > 1$ samples, but selecting an appropriate set of individuals to aggregate for analysis can be complicated. Unrecognized spatial structure is one reason for this, as aggregating across a broad spatial scale may produce a Wahlund effect that biases inbreeding estimates (70), and disentangling true inbreeding estimates from this bias is difficult (71, 72). This is demonstrated in **Figure 1*d***, where two weakly differentiated populations ($F_{ST} = 0.012$) each display no sign of inbreeding when analyzed independently ($F_{IS} \approx 0$), but a sample taken from across the two populations appears inbred ($F_{IS} = 0.12$). Interestingly, H_O is also expected to vary spatially across a population's range, with higher H_O at the center of the distribution and lower H_O at the edges (70). This spatial variability in H_O is not due to bias, as these estimates reflect genuine variation in individual heterozygosity, but it could lead to biased comparisons if populations are sampled with different strategies. In some cases, inbreeding may be better estimated through individual heterozygosity or ROH (68, 72). Conditions under which individuals should be aggregated for estimating inbreeding through F_{IS} remain to be defined.

One exciting development in conservation genomics is the potential to assess heterozygosity in older samples, such as from museums. These are often compared against recently collected samples to test for losses in genetic variation and adaptive capacity. However, data from older samples must be filtered carefully, as DNA degradation can increase the frequency of missing data, which is associated with errors. **Figure 2** presents a reanalysis of SNP heterozygosity estimates to highlight how filtering and display choices can affect interpretation. **Figure 2*a*** presents heterozygosity from unfiltered SNP genotypes, where samples from 1997 to 2001 are compared against those from 2011 to 2017. Individual heterozygosities are summarized as point estimates of means and standard errors, and a linear regression line describes the temporal trend. Reducing data to point estimates as per **Figure 2*a*** can obscure how variation is distributed in the data set, while showing the individual heterozygosity estimates as in **Figure 2*b*** can help illustrate the distribution of variation. **Figure 2*c*** shows the relationship between individual missing data and heterozygosity, and **Figure 2*d*** presents the original data filtered to remove all sites with any missing genotypes. Samples from 1997–2001 had 25% higher missing data than those from 2011–2017, and heterozygosity was positively correlated with missing data in both 1997–2001 and 2011–2017. After filtering missing data, the original decline in heterozygosity is much less pronounced (cf 73). Although this may not be wholly due to errors in the older samples, it does reiterate issues around how missing data are treated.

5. GENOMIC APPROACHES HELP IDENTIFY KINSHIP AND RELATEDNESS IN CONSERVATION

Genetically, kinship (or relatedness) has two measures, continuous and categorical. Continuous kinship measures the degree of genetic similarity, which is taken as evidence of identity by descent. Continuous kinship is useful in captive breeding programs, where relatedness of founding

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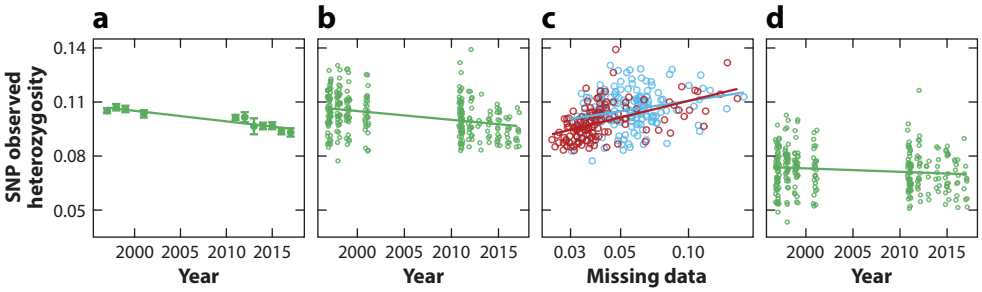

Figure 2

Single-nucleotide polymorphism (SNP) heterozygosity with different presentation and filtering. (*a*) Heterozygosity with no additional filtering, as mean ± standard error. (*b*) The same data as in panel *a* presented as individual heterozygosities. (*c*) The relationship between individual missing data (log-scaled) and heterozygosity; blue points represent 1997–2001, red points represent 2011–2017. (*d*) The same data as in panel *b*, filtered to remove sites with any missing genotypes. Missing data: 1997–2001 = 5.7%; 2011–2017 = 4.6% (t-Stat$_{285}$ = 4.1, $P = 5.4 \times 10^{-5}$). Regressions: (*a*,*b*) ($R^2 = 0.11$, $F_{1,285} = 35.1$, $P = 8.9 \times 10^{-9}$); (*c*) 1997–2001 ($R^2 = 0.05$, $F_{1,156} = 7.7$, $P = 0.006$), 2011–2017 ($R^2 = 0.23$, $F_{1,129} = 38.3$, $P = 7.5 \times 10^{-9}$); (*d*) ($R^2 = 0.02$, $F_{1,285} = 4.6$, $P = 0.034$). Data come from Reference 73, supporting data 4 (**https://doi.org/10.26180/5ef6ec4550f3b**).

individuals may be unknown (74), and can be used to fit genomic animal models for estimating trait heritability (see Section 7). Categorical kinship is used to assign dyads to specific categories of relatedness, such as parent–offspring or sib–sib. Assignment to relatedness categories is challenging but can have a wide range of applications to conservation, such as estimating population sizes (75), dispersal rates (76), dispersal barriers (50, 77, 78), landscape resistance to gene flow (51, 79), movement direction (79, 80), sexual bias in philopatry (81), and local recruitment (82).

Critically for conservation, because close kin are separated by only a handful of meiotic events, information encoded between close kin reflects equivalently recent processes. This can enable rapid assessment of demographic changes caused by habitat fragmentation or other disturbance. For spatial data, these fine temporal scales translate into spatial scales covering the past few generations of dispersal. Close kin data can thus detect partial dispersal barriers operating at the scale of individual movement (78) and have greater power for detecting recent dispersal patterns than analyses that consider the entire population (50). In spatial population genetics theory, closely related individuals occupy nodes that rapidly coalesce on a spatial pedigree (83). Future integration of spatial population genetics theory into kinship-based conservation work should provide useful insights, such as how local kinship structure relates to dispersal, heterozygosity, and population density (83).

Continuous kinship can be quantified through empirical genetic distance or distances derived from principal components. For categorical kinship, the simplest method is to base inferences on relatedness thresholds using estimates of continuous kinship. For more precise estimates that differentiate among relationships within kinship orders (e.g., parent–offspring and sib–sib within first-order kin), methods can incorporate expected proportions of shared alleles [e.g., PC-Relate (84)] or base inferences on homologous microhaplotypes (85) or the number and size of shared identity by descent segments [e.g., ibis (86)]. Methods may also seek to estimate the likelihood of the entire pedigree rather than evaluating each pairwise relationship independently [e.g., sequoia (87)]. Pedigree-based methods may help evade issues of impossible relationships (e.g., a triad of two full-sib pairs and a half-sib pair), but comprehensive comparisons of pedigree and pairwise

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methods are currently lacking. A recent evaluation of pairwise methods applied to captive-bred populations indicated considerable heterogeneity in performance (88).

Although a full evaluation of methods is beyond the scope of this review, some general considerations are relevant to conservation. Firstly, methods that detect close kin from their genetic similarity relative to a background of unrelated individuals require an accurate measure of allele frequencies that may be unavailable for rare taxa. Secondly, population inbreeding and admixture need to be considered, which respectively lead to higher or lower genetic similarity among individuals than expected from their genealogical ancestry; some methods appear to outperform others in the presence of inbreeding (87) and admixture (86). Thirdly, reference genome quality is important, as methods using identity-by-descent segments require contiguous assemblies that may be unavailable for understudied taxa. Finally, to what specificity do kin need to be categorized? If the aim is to identify genetically similar individuals for removal from an analysis (e.g., for GWA), then the process is trivial. For detecting recent movement across barriers or between habitats, it may be sufficient to identify dyads to either the first or second order. However, specific categorizations are needed for assessing sex-biased philopatry (81) or using hierarchical kin distributions to model intergenerational dispersal (76).

6. ASSESSING GENETIC HEALTH USING PROXIES OF GENETIC LOAD

The impact of deleterious mutations on population fitness is the genetic load of a population. In conservation, genetic load typically is divided into an inbreeding or segregation load, due to expression of deleterious alleles from matings between related individuals, and drift load, due to the accumulation of deleterious alleles in small populations (summarized in 1). In small populations, a primary expectation is that genetic load will contribute to a state of mutational meltdown (89), which is also caused by weakened purifying selection and elevated genetic drift, and these impact responses to selection (90, 91). Investigating the molecular mechanisms underpinning genetic load has been challenging historically, and studies have often relied on phenotypic measures of fitness traits and hybrid vigor or genetic data from a few (typically microsatellite) loci as evidence of inbreeding depression (92).

Although actual estimates of genetic load require knowledge of the dominance and selection coefficients of deleterious mutations, genomic approaches have assisted in identifying potentially deleterious mutations without directly measuring fitness. Rather, these approaches aim to identify genomic regions contributing minimally to load due to strong purifying selection, or to use known gene models to predict how local mutations affect load (90, 91). These genomic proxies for genetic load may reflect the frequencies of deleterious mutations or homozygous genotypes. This corresponds to the realized or expressed load, which is the component of genetic load reducing fitness in the current generation, while a masked component may also be expressed in future generations (90). Small population sizes are expected to have a low level of masked genetic load, whereas their realized genetic load increases as deleterious alleles become fixed.

Recent studies have investigated how demographic histories have shaped different contemporary genetic loads in wild populations. In a comparison of two gorilla subspecies, historical and extant populations of the endangered *Gorilla beringei beringei* had indistinguishable genetic loads, whereas extant populations of the critically endangered *Gorilla beringei graueri* had significantly higher genetic loads than historical populations, presumably due to large population declines in recent decades (64). Similarly, in the Montezuma quail, *Cyrtonyx montezumae*, many deleterious alleles have evolved recently and therefore have had insufficient time to be purged from smaller populations (93). Population simulations can predict the detrimental effects of genetic load in wild organisms (e.g., 53, 54). For instance, in Montezuma quail, historically large populations

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were predicted to suffer a greater reduction in individual fitness due to load if there were severe population declines (54). If a well-annotated reference genome is available, mutations in protein-coding genes could provide insights into the molecular mechanisms of inbreeding depression and mutational meltdown, as illustrated in the kākāpō (*Strigops habroptilus*) (94).

One underexplored use of genomic data is to use genetic load proxies as metrics for monitoring conservation efforts in real time (95); inferred deleterious mutations are effectively candidate loci of genetic load. In a genomic monitoring approach on a reestablished population of the gray wolf, *Canis lupus*, the proportion of genotypes homozygous for deleterious mutations increased over a 30-year period (96). This translated into an increased realized load in the reestablished population, suggesting that stable gene flow with external populations should be encouraged for long-term viability. Genetic load proxies could also help evaluate the outcomes and trajectories of genetic rescue attempts and could facilitate captive breeding programs by identifying pairs whose offspring should have reduced load. For instance, in Chinese pangolins (*Manis pentadactyla*), several hundred loss-of-function mutations were identified across three populations (97). By considering the number of these mutations that would be introduced in translocations across these populations, population translocations generating the lowest genetic load could be identified. However, genetic load proxies do not necessarily correlate directly with the endangerment of species because genetic load has many components. Indeed, threatened species do not necessarily have more estimated load than non-threatened species (98).

Despite their potential to provide insights into genetic health, genetic load proxies have several limitations, as discussed elsewhere (1, 90, 91). We highlight several here. First, there are no standardized, user-friendly pipelines for identifying deleterious mutations and calculating genetic load proxies. The bioinformatic complexity of such analyses may be prohibitive for someone without a strong bioinformatics background. Second, empirical evidence on causal associations between genetic load proxies and fitness remains scarce. In the absence of manipulative experiments, studies on wild organisms could conceivably test for this if they have both genotypic and phenotypic data, treating individual fitness as a function of individual genetic load. Third, we need empirical data on how environmental factors influence the expression of genetic load, given that deleterious effects can be environmentally dependent (99), by treating fitness as a function of genetic load, environment, and their interaction. Finally, estimates of dominance and selection coefficients would greatly facilitate the modeling and predictive capacities of genetic load studies. This is hard because of the requirement to understand phenotypic effects of deleterious mutations in the wild.

Candidate loci: loci thought to have functional importance, typically affecting traits related to adaptation and fitness, typically identified through GWA and GEAs to identify adaptive genetic variation; modern genomic methods can also identify loci potentially contributing to genetic load

Genetic architecture of adaptation: description of the number, type, genomic distribution, and effect size of genes contributing to adaptation

Adaptive markers: genetic markers thought to be adaptive, with effects on traits related to fitness under particular environmental conditions

Genomic offsets: projected evolutionary changes required for contemporary populations to adapt to future environmental conditions (similar to genomic vulnerability); estimates of required degree of adaptation typically are based on contemporary patterns of genotype–environment associations

7. INVESTIGATING ADAPTATION: POPULATION GENOMIC VERSUS QUANTITATIVE GENOMIC APPROACHES

In CG there is currently much interest in characterizing the genetic architecture of adaptation. Candidate loci of adaptation can be identified via GWAs and GEAs, which link genotypes to variation in adaptive traits or environmental factors, respectively (**Figure 3*a*,*b***). Management actions can be built around insights from functionally important genetic variation at adaptive markers. For example, translocations, breeding programs, and releases can incorporate knowledge of where specific genotypes are likely to be adaptive or maladaptive (8). Predictive models can assess which populations are most at risk from future environmental shifts, such as those estimating genomic offsets (or genomic vulnerability) under climate change (63, 100, 101). Genomic monitoring programs can also target candidate adaptive markers (95). Discussion regarding using genome editing in conservation programs to artificially introduce adaptive genetic variants is growing (102).

Despite the value of association studies for studying adaptation, there are limitations to these methods, particularly in the context of wildlife (reviewed in 4, 6, 7). An overarching concern is whether outlier loci are truly biologically relevant. Many factors can lead to biases or

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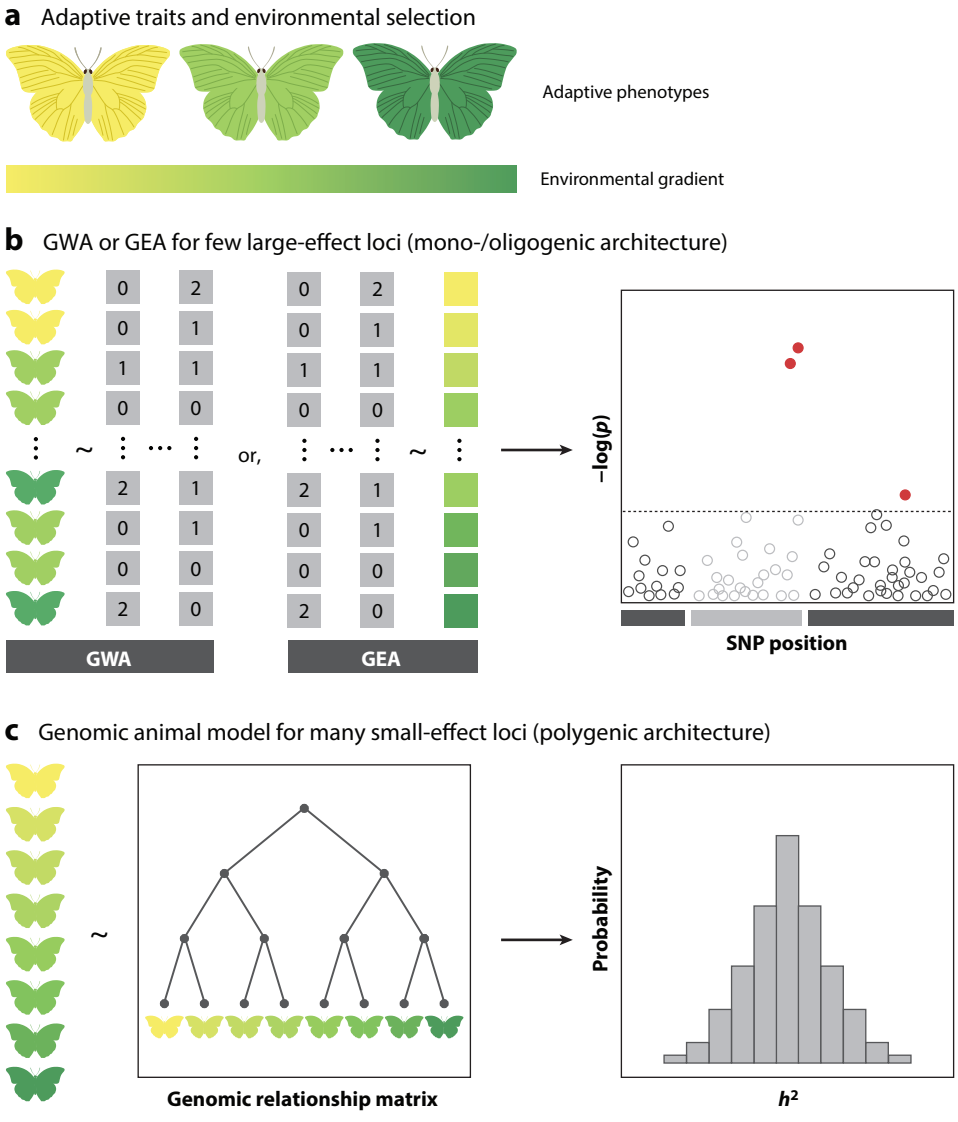

Figure 3

Schematic overview of alternative pathways for studying adaptation with genomics. (*a*) A trait hypothesized to be adaptive, or environmental gradient(s) hypothesized to drive selection, is to be measured. Here, an imaginary butterfly species with a range of color morphs from yellow to dark green (adaptive phenotypes) that offer camouflage to different microhabitats (environmental gradient). (*b*) GWA or GEA to detect loci with mono-/oligogenic architectures. In a GWA, phenotypes are modeled as a function of genotypes, whereas in GEA, genotypes are modeled as a function of environment. Outlier loci with scores significantly above a genome-wide threshold are identified as candidate loci of adaptation. (*c*) Genomic animal models to infer the heritability of a trait with a polygenic architecture. Phenotypes are modeled as a function of additive genetic breeding values, which are estimated from a genomic relationship matrix. The relative contribution of V_A to the V_P can be used to infer h^2 and its probability distribution for a sample. This does not require many individuals, provided there is variability in genetic relationships and these relationships can be estimated accurately. Abbreviations: GEA, genotype–environment association; GWA, genome-wide association; h^2, narrow-sense heritability; SNP, single-nucleotide polymorphism; V_A, additive genetic variance; V_P, phenotypic variance.

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misidentification of adaptive signatures in genetic markers (4, 103), and establishing consistent connections between genes and phenotypes remains difficult in conservation (2). Incorrect spatiotemporal scales for measuring adaptation and contextual changes in the genetic architecture of adaptation (i.e., genotype-by-environment interactions) further complicate this issue (104, 105). GWAs and GEAs are also biased toward detecting large-effect loci, so they are more likely to accurately detect monogenic and oligogenic architectures over polygenic architectures composed of many small-effect loci. This is problematic because many adaptive traits are polygenic, including traits likely to be important in a changing world (106, 107). Still, methodological advances are helping overcome difficulties associated with polygenicity (108), paving a way for testing the predictive capacity of candidate loci and using GWAs and GEAs to study polygenic adaptation.

Large-effect loci: genetic loci that contribute to a large proportion of the variance in a trait or adaptive response

Monogenic: relating to a single gene, e.g., traits with a monogenic architecture

Oligogenic: relating to several genes, e.g., an oligogenic architecture

Polygenic: relating to many genes, e.g., a polygenic architecture

Small-effect loci: genetic loci that contribute to a small proportion of the variance in a trait or adaptive response

Narrow-sense heritability (h^2): proportion of phenotypic variance specifically composed of additive genetic variance, in contrast to broad-sense heritability (H^2), which considers all components of genetic variance (including dominance and epistatic genetic variance)

Genomic predictions are important for validating the biological relevance of candidate loci. This validation may be statistical or experimental in nature. For statistical validation, a training data set is used to fit a GWA or GEA, and the resulting candidate loci provide predictions for a testing data set. Ideally, multiple independent testing data sets are used to validate predictions, although this may be infeasible for many wild organisms. An alternative is to cross-validate the data set with multiple random permutations of training and testing partitions. This was demonstrated in a sample of 213 *Acropora millepora* corals, where 100 jackknifed cross-validations with an 85%:15% training:testing split were used to conclude that polygenic score estimates described 2.6% of the variation in bleaching susceptibility phenotypes (109).

Experimental associations between phenotype and candidate loci can also validate genomic predictions. In the marine copepod *Eurytemora affinis*, experimental selection to different salinity levels helped identify outlier loci and estimate polygenic scores for wild populations. These putatively adaptive makers showed significant relationships with field salinity levels, suggesting that loci identified under experimental conditions were ecologically relevant (110). Common gardens provide another route for experimental validation, although this is more commonly used in plants (e.g., 111). Such experiments might be particularly useful for ground-truthing genomic offsets estimated under climate change by moving genotypes to common gardens with climates that mirror those predicted in the future (7). Nonetheless, this approach may rarely be possible for species of conservation concern.

The choice between statistical or experimental validation of candidate adaptive markers is species dependent. Experimental manipulation in species of conservation concern will be limited in power by small population sizes, permit requirements, and the inability to sacrifice individuals to measure certain adaptive traits (e.g., thermal tolerance). Many species are also recalcitrant to experimental manipulations. Whereas small animals are amenable to laboratory experiments, and sedentary animals are well-suited for common gardens, large, mobile animals are suited to neither. Characteristics of at-risk species also impact suitability for GWAs and GEAs. Without large sample sizes, there is little statistical power to test for genotype–phenotype associations in GWAs, precluding their use in rare species or those difficult to sample (112). GEAs can be more feasible, provided that an adequate number of populations are sampled along meaningful environmental gradients. These studies are strengthened by reference genomes that can place loci in their genomic context (see Section 2).

The quest to characterize the genetic architecture of adaptation has detracted somewhat from alternate quantitative genomics approaches for studying adaptation. A key aim in such studies is to estimate the narrow-sense heritability (h^2, herein heritability) of traits, that is, the proportion of phenotypic variance (V_P) composed of additive genetic variance (V_A). Heritability dictates the response to selection and can provide insight into a population's adaptive capacity; h^2 is therefore an important parameter for modeling evolution (113). Estimates of h^2 in wild organisms have been obtained through use of animal models (114). These mixed-effects models allow partitioning of

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both fixed and random effects contributing to V_P. Pedigrees are used to derive genetic relationship matrices that control for the additive genetic covariance among individuals, which enables estimation of V_A. Absent a pedigree, genomic animal models (GAMs) can be fitted by deriving genomic relationship matrices from genotype data that act as surrogates for pedigree-based genetic relationship matrices (**Figure 3*c***).

Genetic (genomic) relationship matrix: matrix describing the pairwise relatedness among individuals using discrete values derived from pedigrees (genetic relationship matrix) or continuous values derived from genome-wide genetic markers (genomic relationship matrix)

GAMs provide a bridge between population genomic and quantitative genetic investigations. Any type of genomic data (including RADseq) could be used to infer genomic relationship matrices, and GAMs do not require reference genomes because the effects of individual loci are not investigated. This makes GAMs attractive when adaptive traits are polygenic (115). They are also relatively straightforward to implement (116). Despite these advantages, GAMs are not widely used in studies of wild organisms, possibly because molecular ecologists have recently been motivated by identifying genes and candidate loci of adaptation (but see, for example, 117–119). However, power to estimate V_A and h^2 with a GAM does depend on sample sizes and structure of the genomic relationship matrix (114, 115). If there is sufficient variation in relatedness among individuals, large sample sizes are not required. Therefore, unlike GWA and GEA methods, GAMs can estimate h^2 in species that are rare and have small population sizes (115).

To summarize, genomics can be used to investigate adaptation in conservation through two alternate routes, one population genomic (using GWAs and GEAs; **Figure 3*b***) and another quantitative genomic (using GAMs; **Figure 3*c***). Both provide useful genetic information for modeling evolution but vary in terms of the exact insights they provide and specifics of their limitations. Researchers should be realistic about insights into adaptive processes that they can expect from their study species, and within the limits of their budget and study design. Candidate adaptive makers from GWAs and GEAs need validation to increase confidence that results are biologically relevant.

8. MANAGING WILD OR CAPTIVE POPULATIONS

Genomic data are becoming easier and cheaper to collect and process, adding to a diverse and rapidly expanding toolbox of resources for conserving species and managing biodiversity. Some conservation programs now have access to increasingly detailed genomic information, via high-quality reference genomes, and better ways of extracting genomic data from noninvasive sources [e.g., kākāpō (120)]. Nevertheless, management decisions for many threatened species are still made without much reference to genetic or genomic data (121).

8.1. Management Applications

Genomic approaches are being used currently for a variety of tasks, such as estimating relatedness in pedigrees (88), characterizing introgression of climate-adapted alleles (63), characterizing levels of genetic variation across populations (122), and delineating taxa and establishing patterns of gene flow (123). These approaches build on key past findings in CG (1, 124, 125) but also provide opportunities to explore new issues in wildlife conservation. Nevertheless, as genomic methods become increasingly sophisticated, there is an increased risk of genomic approaches being seen as too difficult to apply in a practical sense, or as being applied inappropriately when appropriate bioinformatic expertise is not available and when there is a poor understanding of underlying theory (126).

Decisions on investing funds into genomic approaches should be based on how much additional value can be gained from genomic data, especially when resources for conservation are limited and other genetic information has been collected previously. Sometimes genomic data provide little additional information over and above what was established previously from other markers, as in

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the case of introgression in threatened black stilts (*Himantopus himantopus*), where a RADseq study produced the same finding as that established previously from microsatellite markers (127). However, in other cases, completely different pictures have emerged when genomic data are employed, such as in comparisons of mitochondrial DNA variation and genome-wide SNPs in the wingless grasshopper, *Vandiemenella viatical* (128), and in the highly vagile Galapagos shark, *Carcharhinus galapagensis* (129). Genomic data provide a more dynamic perspective for defining conservation units for such species (123), which should aid their management.

Neutral markers: genetic markers presumed to have no effects on fitness; in practice, can be difficult to define because of linkage with markers under selection and because markers may be neutral under only some environmental conditions

Genomic approaches make it easier for managers to standardize measures of population differentiation and genetic variation (Sections 3 and 4). For assessing autosomal heterozygosity, we advocate for methods that help standardize estimates for comparisons across studies or across time and have included code for this (**Figure 1**). Because bioinformatic pipelines are complex and genomic resources (e.g., reference genomes) vary in quality, we encourage that all raw data (raw sequences, not only the called SNPs) be made available with published studies. Appropriate archiving of raw data will facilitate reanalyses and combining old and new data in future investigations. As well as heterozygosity, genomic data can provide definitive measures of adaptive potential in populations, helping guide management decisions about whether to conserve a species in situ or consider more drastic action in cases where there is little genetic variation and populations are extremely restricted in size [e.g., sponges (130)].

Population declines can be tracked by monitoring genomic variation across time, such as in northern quolls (*Dasyurus hallucatus*), whose populations were decimated following the arrival of cane toads (131). Genomic monitoring can help assess genetic health after management actions, such as the tracking of genomic diversity when quolls were moved from mainland to island populations (131), and in the maintenance of genomic diversity in captive versus wild populations of Kaua′i honeycreepers (*Oreomystis bairdi* and *Loxops caeruleirostris*) (132). Genomic data can also track the success of restoration programs, such as in the southern pygmy perch (*Nannoperca australis*), where low inbreeding and persistent genetic variation were documented in restored populations (133). However, as emphasized in this review, there are pitfalls in using genomic data for monitoring changes in population health if potential confounding effects are not considered, such as those associated with DNA quality in old samples (**Figure 2**).

Apart from tracking overall genetic variation, other useful population genomic measures for managers include inbreeding coefficients (F_{IS}), ROH, and measures of genetic load. ROH have been analyzed to identify differences in inbreeding across feral horse populations (134), which may guide future genetic mixing strategies. However, when using F_{IS} to assess inbreeding, population structure must be considered (Section 4). Changes in genetic load can also be tracked [e.g., leopards (135)], but there remain challenges in separating different components of genetic load (Section 6). In addition, heterozygosity–fitness correlations can be used to estimate inbreeding effects (136), but these require appropriate model fitting and handling of outlying individuals to ensure results are not biased.

8.2. Adaptive Markers

Management guidelines have been developed for establishing patterns of population differentiation and genetic variability using putative neutral markers specifically (1, 137, 138), yet variation at neutral markers may also be shaped by linkage to markers under selection. In experimental and longitudinal studies, measures of overall levels of genomic variation can be effective in predicting adaptive potential (e.g., 139). High levels of individual heterozygosity can also indicate individuals with higher fitness under stressful conditions, as in frogs affected by chytrid fungus (140). Nevertheless, some authors have advocated a greater focus on adaptive variation in making management decisions (e.g., 141). For wildlife, this remains challenging in part due to the difficulty

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of predicting polygenic selection responses in populations using adaptive markers, and predicting which markers will contribute to future adaptation when future selection pressures are unclear or unpredictable (1, 2, 112, 138). Responses to most environmental changes in the future will likely be polygenic; even responses to diseases can be complex due to the involvement of multiple disease resistance loci (142) and the importance of diverse selection responses, such as the shift in Tasmanian devils to earlier breeding (143).

Adaptive markers nevertheless remain a useful tool in some management contexts, provided they are applied with caution. Applications include using genomic monitoring to track changes in phenotypes involved in responding to environmental threat, as undertaken in the coral *A. millepora* when exposed to bleaching events (109). Adaptive markers have also helped identify past adaptive shifts in populations through GEA analyses (Section 7), which can inform genetic mixing strategies like climate-based provenancing, an increasingly considered strategy in biodiversity conservation (63, 144).

8.3. Prioritizing Populations

Genomic approaches have value in delineating population groupings and understanding processes that have shaped them (Section 3). These insights can be integrated with other methods, such as species distribution modeling [e.g., for quolls (145)] to assist in forward planning, particularly when populations still persist at a reasonable size (8, 9). Overall levels of genetic variation will likely remain the most important parameter when prioritizing populations for conservation. The genetic uniqueness of populations will also be important, although this can conflict with overall genomic variation if genetic drift has generated population uniqueness (52).

Adaptive markers could provide a second measure of the importance of genetic variation in cases where multiple populations are available for evaluation. One proposed measure is the population adaptive index (146), in which adaptive loci are identified via outlier approaches and populations are scored by the percentage of loci with allelic frequencies significantly different from other populations. This method aims to maximize the amount of unique genetic variation available for future adaptative potential but is less focused on understanding maladaptation to future conditions. Alternatively, methods like genomic offsets are geared toward inferring future maladaptation (101) and may help identify populations requiring the greatest future adaptive shifts, assisting management priorities.

There are challenges in establishing GEAs and genomic offsets (Section 7). Additionally, if GEAs are used to infer future adaptive changes, managers need to be confident in their modeling of past and future environments and other genetic processes affecting population persistence. For example, genomic vulnerability estimates in the yellow warbler (*Setophaga petechia*) were proposed to exhibit associations with expected population declines (100). However, a criticism of this work is that the modeling of climatic adaptation and projected mismatch to future environments failed to accurately account for historical population trends, climate, and potentially confounding effects of neutral population structure (147). Validation of candidate loci through genomic predictions remains key to future adaptive evolutionary investigations in conservation but is difficult for wildlife. Careful consideration of filtering parameters and study design will help maximize power and reduce false positives (4, 103).

8.4. A Potential Roadmap for Wildlife Genomics

We conclude with some suggested workflows and resulting actions that managers may take in applying wildlife conservation genomics. **Figure 4** summarizes processes that can be informed by genomic approaches and the types of management actions that might be applied. These include

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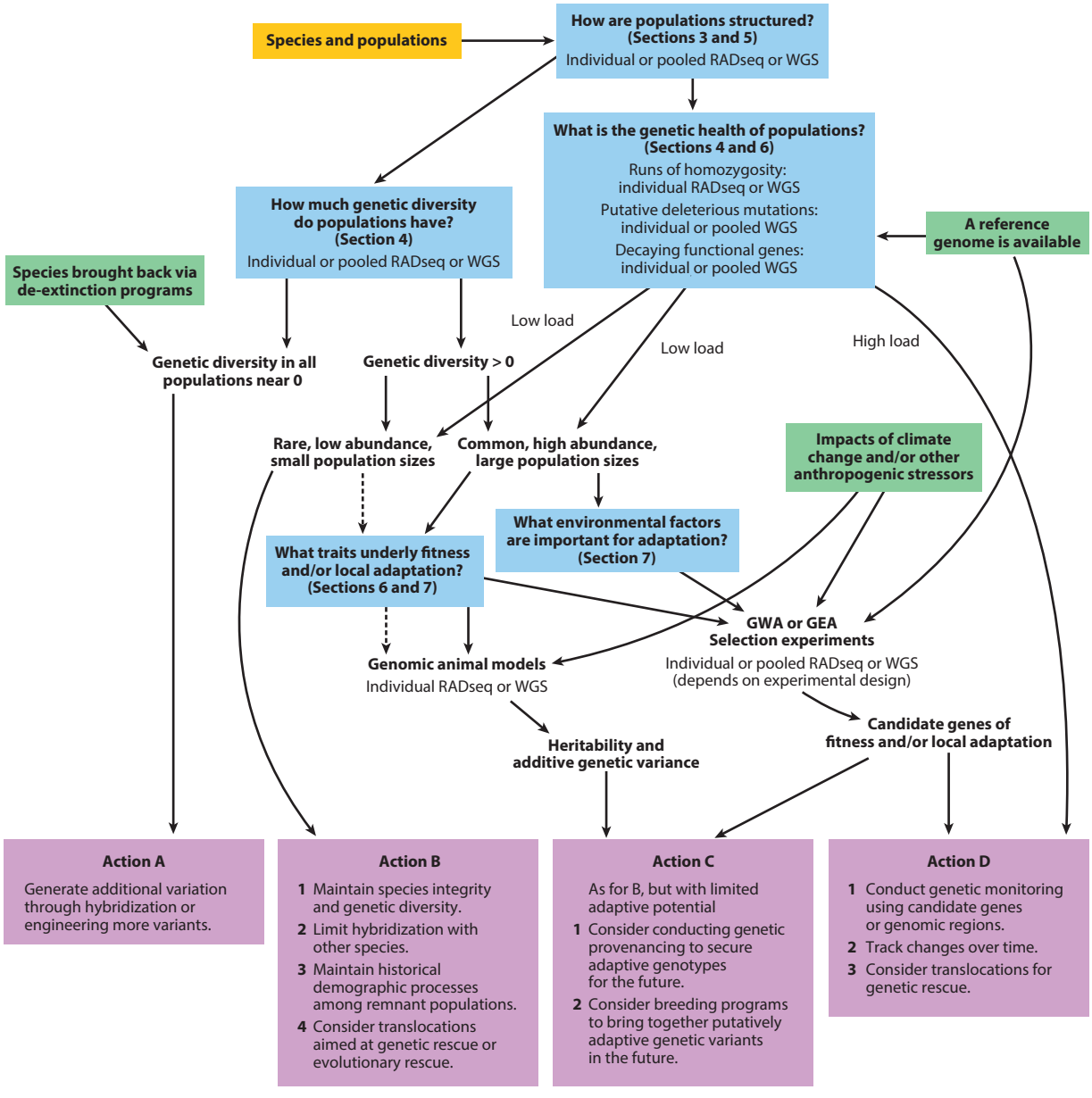

Figure 4

Framework for integrating genomic data into wildlife conservation. All conservation studies begin with one or more focal species and a set of sampled populations (*yellow box*). From there, a series of key questions are needed to define the goals of a study and the genomic data to collect (*blue boxes*). External factors may affect the decisions being made (*green boxes*). Ultimately, different conservation actions will be reached depending on the biology, budget, and experimental design implemented in a conservation project (*purple boxes*). Abbreviations: GEA, genotype–environment association; GWA, genome-wide association; WGS, whole-genome sequencing.

undertaking drastic actions to develop genomic variation in threatened populations (Action A) through to focusing on maintaining existing levels of variation (Action B), considering additional options to increase adaptive capacity (Action C), and planning translocations to decrease inbreeding load (Action D). Genomic approaches help in making these decisions, but ultimately a host of nongenetic factors will also need to be considered. Although the array of options for incorporating genomics in management can be daunting, the myriad potential insights and the ongoing drop in genomic processing costs should help in (cautiously) increasing applications of this valuable toolbox.

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SUMMARY POINTS

1. Decisions to invest in a reference genome should be based on study objectives. Declining costs and technological advances mean that genomes up to 6 Gb could be assembled relatively affordably.
2. Genetic structure can be detected with more power and precision than ever before, but filtering and modeling must be done carefully to ensure interpretations of structure are meaningful to conservation.
3. Heterozygosity varies across the genome, and these genomic patterns are worth investigating. Filtering choices strongly affect estimates of overall genome-wide variation.
4. Pairwise kinship is a useful means of assessing recent changes to populations, and genomic data allow for categorically different approaches than used previously.
5. Genetic load proxies provide opportunities to track the genetic health of populations, although there are lingering issues around linking genomic estimates of genetic load with fitness in the wild.
6. Population genomics and quantitative genomics provide two alternate routes for studying adaptive evolution. Each provides different insights and is suitable under different biological contexts, budgetary constraints, and experimental designs. Validating the biological relevancy of candidate loci of adaptation is an ongoing challenge.
7. It remains challenging to prioritize populations for conservation based on adaptive markers identified from genomic approaches as opposed to overall levels of genetic variation.
8. Successful implementation of genomic data in management strategies requires selecting the right approaches, which depends on the species of concern, how much money and time can be devoted to genomics work, and available bioinformatic resources and training.

DISCLOSURE STATEMENT

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